

Week 4 Answers — The Amdahl Reality Gap

Empirical speedup fitting, false sharing, and loop scheduling in OpenMP.

This answersheet uses the completed laptop run in [runs/20260924_121506_410](https://github.com/ros1nant/20260924_121506_410). All reported benchmark times come from that session; diagnostic profile timings are identified separately.

1. Student and machine information

Required field	Recorded information
Student full name	Ainabek Aisultan
Student ID	230103254
Git URL	https://github.com/ros1nant
Practicum date	24 September 2026
Recorded benchmark interval	07:15:06–07:15:38 UTC; 12:15:06–12:15:38 at UTC+05:00
Laptop	ASUS ROG Strix G614JV_G614JV
Operating system	Windows 11 Home Single Language, 64-bit, version 10.0.26200
CPU and generation	Intel Core i7-13650HX, 13th generation, Raptor Lake
Physical cores	14: 6 Performance-cores and 8 Efficient-cores
Logical threads	20; <code>omp_get_max_threads()</code> and <code>omp_get_num_procs()</code> both report 20
L1 / L2 cache line size	64 bytes, as reported by the Windows cache query
Installed memory modules	Two 8 GiB modules; reported configured speed 4800
Compiler	GCC 16.1.0, MSYS2 build
Compilation flags	<code>-O2 -fopenmp -std=c11 -Wall -Wextra</code>
OpenMP configuration	<code>OMP_DYNAMIC=FALSE, OMP_NUM_THREADS=20</code> ; default runtime affinity
Target workload N	13,254,000 starting integers

The laptop details and cache line sizes come from [hw_info.txt](#). The P/E breakdown and generation are confirmed by [Intel's i7-13650HX specifications](#). Power mode, temperatures, background activity, and the full practicum duration were not recorded. The benchmark interval above does not establish the duration of the whole lab session.

2. Workload derivation and correctness

Student ID **230103254** ends in **3254**. The original benchmark was invoked with 3254, so the raw logs retain that entered value. Both inputs produce the same workload because the program uses only the last four digits:

$$\begin{aligned} N &= 10,000,000 + (3254 \times 1,000) \\ &= 13,254,000 \end{aligned}$$

For each integer i from 1 through N , the kernel counts the steps to reach 1: divide an even value by 2; replace an odd value with $3 \cdot n + 1$. The program finds the maximum stopping time, sums the stopping times modulo 1,000,000,007, and counts how many starting integers require more than 100 steps.

Verification result	Value
Maximum stopping time	688
Checksum modulo 1,000,000,007	96,515,714
Starting integers requiring more than 100 steps	10,684,763
Raw benchmark repetitions	45 across 15 configurations
Agreement between sequential and parallel results	All 45 repetitions match
Requested versus actual OpenMP thread counts	Match in every recorded repetition

The separate scheduling profiles each account for all **13,254,000** iterations and **2,096,515,728** total Collatz steps. This independently checks the profile totals against the recorded checksum:

$$2,096,515,728 \bmod 1,000,000,007 = 96,515,714$$

[session.json](#) reports `complete`. Its recorded SHA-256 hashes match the source and local automation files used for this run. The three C source/header files are included in this submission; the runner and analysis helpers remain in the local working folder. The [console log](#) also records successful completion and agreement with the sequential baseline.

3. Measurement method and sequential baseline

Each configuration was executed three times. Run 1 was discarded as the warm-up; the arithmetic mean of runs 2 and 3 is the reported time. Timings use `omp_get_wtime()`. The first run was not preceded by an explicit cache flush, so the worksheet's cold-run label does not establish that every cache was cold.

The runtime reports `omp_get_wtick() = 0.001 s`: **1 ms resolution**. Printing nine decimal places does not establish nanosecond or sub-microsecond precision. Times below are rounded to six decimal places for readability; the original values remain in [results.csv](#).

Sequential Run 1, discarded (s)	Run 2 (s)	Run 3 (s)	T_seq (s)
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	2.039000	1.989000	1.984000	1.986500
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$$T_{seq} = (1.989000082 + 1.983999968) / 2$$

$$= 1.986500025 \text{ s}$$

The true sequential loop is distinct from the one-thread OpenMP loop. Empirical speedup uses **T_seq**, not the OpenMP one-thread time. Profile bookkeeping runs were performed separately and are excluded from all benchmark averages.

4. Table 1 — Scaling and the Amdahl fit

$$T_k = (\text{Run 2} + \text{Run 3}) / 2$$

$$S_{emp}(k) = T_{seq} / T_k$$

$$S_{emp}(2) = 1.986500025 / 1.488999963$$

$$= 1.334117$$

$$1 / S(2) = (1 - p) + p/2 = 1 - p/2$$

$$p = 2 \times (1 - 1/S_{emp}(2))$$

$$= 0.500881003$$

$$= 50.0881003\%$$

$$S_{theo}(k) = 1 / [(1 - p) + p/k]$$

$$\Delta(k) = S_{theo}(k) - S_{emp}(k)$$

$$\text{Linear ideal: } S(k) = k$$

Threads k	Run 1, discarded (s)	Run 2 (s)	Run 3 (s)	Average T_k (s)	S_emp(k)	S_theo(k)	Delta(k)
1	2.803000	2.814000	2.800000	2.807000	0.707695	1.000000	0.292305
2	1.482000	1.491000	1.487000	1.489000	1.334117	1.334117	0.000000
4	0.791000	0.796000	0.773000	0.784500	2.532186	1.601693	-0.930493
8	0.443000	0.414000	0.446000	0.430000	4.619768	1.780217	-2.839551
14	0.296000	0.280000	0.288000	0.284000	6.994717	1.869522	-5.125196
16	0.267000	0.267000	0.260000	0.263500	7.538900	1.885284	-5.653616
20	0.226000	0.241000	0.259000	0.250000	7.946000	1.907803	-6.038197

The required powers of two are supplemented by **14 physical cores** and **20 logical threads**. This table uses static scheduling throughout. The largest measured speedup in this scaling series is **7.946000× at 20 threads**.

The scalability graph appears on the final page of this PDF.

Negative gaps mean measured speedup is **above** the fitted theoretical curve. The two-thread fit has a formal infinite-thread limit of about **2.00353×**, yet the measured scaling series exceeds it from four threads onward. This shows that the fitted p does not describe a fixed physical serial fraction for this implementation across all thread counts; it is not a violation of Amdahl's law.

5. Table 2 — False sharing

Both variants use static scheduling with **14 threads**, equal to the recorded physical core count. Both return the same maximum, checksum, and hit count. The mitigated implementation uses an OpenMP reduction.

Variant	Threads	Run 1, discarded (s)	Run 2 (s)	Run 3 (s)	Average time (s)	Throughput (iterations/s)	Time / reduction time
Naive adjacent counters	14	0.334000	0.326000	0.335000	0.330500	40,102,873.64	1.167845
OpenMP reduction	14	0.278000	0.287000	0.279000	0.283000	46,833,923.50	1.000000

```
Throughput = N / average time
Penalty ratio = T_naive / T_reduction
               = 0.3305000065 / 0.2829999925
               = 1.167845
```

The naive variant is **16.78% slower** than the reduction variant. Switching to reduction reduces elapsed time by **14.37%**, or approximately **0.0475 s**.

6. Table 3 — Scheduling and workload distribution

All five schedules use **20 threads** and the same reduction-based kernel. The runtime schedule is selected through `omp_set_schedule()` and applied by `schedule(runtime)`.

OpenMP schedule	Chunk size	Run 1, discarded (s)	Run 2 (s)	Run 3 (s)	Average time (s)
static	Default: 662,700 iterations per thread	0.235000	0.233000	0.221000	0.227000
static, 1000	1,000	0.241000	0.229000	0.251000	0.240000
dynamic, 100	100	0.202000	0.197000	0.200000	0.198500
dynamic, 10000	10,000	0.199000	0.200000	0.199000	0.199500
guided	Decreasing chunks; default minimum	0.216000	0.196000	0.211000	0.203500

The following observations use the separate [thread_profiles.csv](#) diagnostic passes. Ranges describe the 20 worker threads within each pass.

Schedule	Iterations per thread, min–max	Active loop time, min–max (s)	Observed distribution
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static	662,700–662,700	0.141–0.229	Equal iteration counts, but total Collatz work varies from 84.27 to 111.53 million steps; completion times differ substantially.
static, 1000	662,000–663,000	0.145–0.250	Round-robin chunks bring work close to 104.19–105.57 million steps per thread, but completion times remain uneven.
dynamic, 100	510,000–819,400	0.200–0.201	Unequal iteration assignments accompany closely grouped completion times.
dynamic, 10000	410,000–960,000	0.215–0.218	Coarser work redistribution also gives closely grouped completion times.
guided	435,245–918,588	0.195–0.223	Adaptive chunks redistribute work, although a longer tail remains in this diagnostic pass.

These measurements describe assigned work and active elapsed time, not direct CPU utilization. Thread IDs were not mapped to P/E cores or SMT siblings. The diagnostic passes are not additional repetitions for Table 3 and must not be averaged into it. Table 3's static result is a separate measurement group from Table 1's static 20-thread result; their averages are retained separately.

7. Q1 — Micro-architectural root cause of false sharing

The naive implementation showed a **1.167845× penalty**, rather than the massive degradation assumed in the question. This CPU's recorded L1/L2 cache line size is **64 bytes**. With 4-byte `int` counters and a 64-byte-aligned array, the 14 active counters occupy 56 bytes within one cache line. Although each worker updates a different counter, workers executing on different cores contend for ownership of that line.

In the **MESI** model, a core must obtain exclusive write ownership and modifies the line into the Modified state. A later writer on another core requires an ownership transfer that invalidates the previous core's copy. Repeated transfers cause coherence requests, invalidation traffic, and stalls on the interconnect (the worksheet's bus traffic). MOESI adds an Owned state, but it also requires single-writer ownership; it does not eliminate false sharing. MESI is used here as the explanatory model, without claiming the logs identify this processor's complete coherence implementation. See [Intel's explanation of coherent cache-line ownership](#).

The reduction accumulates private hit counts and combines them at the end, avoiding repeated shared-line updates. The naive counters are `volatile` to preserve repeated writes under optimization, so the comparison also includes memory-access overhead removed by the reduction. No hardware invalidation counters were collected; the measured penalty cannot be attributed exclusively to cache coherence.

8. Q2 — Hyperthreading, SMT saturation, and physical core ceilings

No, speedup did not continue to grow linearly from 14 to 20 threads. The time fell from **0.284000 s** to **0.250000 s**, and speedup increased from **6.994717×** to **7.946000×**. This is a **13.60% throughput increase** for a **42.86% increase in worker count**. Continuing ideal proportional scaling from the 14-thread point would predict approximately **0.198800 s** at 20 threads, which was not achieved.

The CPU has six P-cores and eight E-cores, with 20 logical threads in total. On this hybrid design, the six additional hardware threads come from the P-cores' SMT capability; they are not six additional physical cores. SMT siblings share execution units, caches, and paths to memory, while retaining their own architectural register state. Consequently, another SMT worker can use otherwise idle capacity but cannot duplicate a core's compute resources. These shared resource properties are described in [Intel's Hyper-Threading documentation](#).

The smaller gain is consistent with shared execution resources, differing P/E core performance, and synchronization or workload imbalance. However, runtime affinity was left at its default, so 14 software threads do not prove one worker was placed on each physical core. These results therefore show diminishing returns at the physical/

logical count boundary, not a controlled isolation of SMT's contribution.

9. Q3 — Empirical versus theoretical Amdahl discrepancy

The required two-thread fit gives $p = 0.500881003$, or **50.0881%**. At eight threads, theoretical speedup is **1.780217×** while measured speedup is **4.619768×**, giving **Delta(8) = -2.839551**. At 16 threads, the values are **1.885284×** and **7.538900×**, giving **Delta(16) = -5.653616**. The fitted curve therefore **underpredicts** measured speedup in this run.

A fixed two-parameter workload split does not capture all differences between the sequential and OpenMP implementations. The one-thread OpenMP time is **2.807000 s**, approximately **41.30% longer** than the sequential baseline. This measured difference shows that an OpenMP execution has costs or execution characteristics absent from the sequential measurement. Deriving p from only the two-thread result folds those effects into an apparent serial fraction; that fraction is not demonstrated to remain constant as more workers are used.

Two physical factors omitted from the simple model are:

1. **Unequal core throughput and placement.** P-cores and E-cores are different execution resources. The mix of cores used, worker migration, and unequal Collatz work can change completion times and barrier waiting as k changes.
2. **SMT resource contention.** Sibling logical threads share a core's execution capacity and caches, so each additional logical worker does not contribute another identical independent processor.

These are plausible contributors given the hardware and profiles, rather than individually measured causes. OpenMP scheduling/reduction overhead and unmeasured clock or power changes can also affect the fit. The data do not establish memory-bandwidth saturation or a literal 49.91% inherently serial algorithm.

10. Q4 — Scheduling trade-offs and queue contention

`dynamic(100)` averaged **0.198500 s**, `dynamic(10000)` averaged **0.199500 s**, and default `static` averaged **0.227000 s**. The dynamic schedules reduce time relative to static by approximately **12.56%** and **12.11%**, respectively. The smallest mean belongs to `dynamic(100)`, but its approximately **1 ms** advantage over `dynamic(10000)` equals the reported timer tick and lies within the observed repetition variation. These two measured averages are too close to establish a reliable performance ranking from two retained runs each.

With $N = 13,254,000$, chunks of 100 create approximately **132,540 chunks**, while chunks of 10,000 create **1,326 chunks**. The smaller chunk requires many more work-distribution requests but can balance uneven work more finely. The profiles show both dynamic policies assigning different numbers of iterations to workers while keeping completion times close. Even though `static(10000)` balances the total number of Collatz steps well, it cannot reassign work from slower workers and averaged **0.240000 s**.

Neither tested dynamic chunk size demonstrated queue-contention overhead outweighing the net benefit of balancing relative to static. Both were faster than static in this session. Therefore, no crossover chunk size can be identified from these data. Queue or lock contention was not directly measured; identifying a threshold would require more chunk sizes, additional repetitions, and suitable profiling. It would be unsupported to claim that chunk size 100 caused a net slowdown here.

11. Source code, reproduction, and submission assets

The complete source is in [collatz.c](#), [collatz_seq.c](#), and [collatz_common.h](#). Both `.c` files contain compilation instructions in their header comments. Compile with

an OpenMP-enabled GCC:

```
gcc -O2 -fopenmp -std=c11 -Wall -Wextra collatz_seq.c -o collatz_seq.exe
gcc -O2 -fopenmp -std=c11 -Wall -Wextra collatz.c -o collatz.exe
```

The local `run_lab.ps1` helper captured hardware, built the programs, ran the correctness self-test, and created the benchmark directory. The local `analyze_results.py` helper derived the tables and plot. These helpers and the setup README are excluded from the submission; the required C source is included. To reproduce the computation manually after compilation, run from this folder:

```
New-Item -ItemType Directory -Path rerun
.\collatz.exe 230103254 14 rerun
```

This uses the full student ID and the recorded physical core count. It writes new raw measurements to `rerun` without replacing the submitted dataset. GCC and its OpenMP runtime must be available in the terminal's `PATH`.

Required deliverable	File included with this answersheet
Hardware specification	hw_info.txt
Full source and compilation instructions	collatz.c , collatz_seq.c , collatz_common.h
Raw benchmark dataset for Tables 1–3	results.csv
High-resolution scalability graph	speedup_plot.png
Written answers Q1–Q4	Sections 7–10 of this file and analysis.pdf
Supporting run metadata and diagnostics	session.json , console_log.txt , thread_profiles.csv

The worksheet explicitly requests `analysis.pdf` or written worksheet answers; `analysis.pdf` is an export of this answersheet. After changing personal details, export it again to keep the PDF and Markdown consistent. Keep this Markdown version in the repository along with its linked run files. The worksheet allows a private repository with instructor access or a `lab1_230103254.tar.gz` archive. Repository access and publishing are separate from the measured results.

Scalability graph - measured, fitted Amdahl, and linear ideal

Amdahl Reality Gap | N=13,254,000

